vllm-sr/vela3-mini

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Name it. Vela finds it.

vLLM Semantic Router × KR Labs · Vela 3.0 family · **vela3-mini** · 4B · base `d3-mini` · Apache-2.0

What it does · Quick start · Evaluation · The family · Collection

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vela3-mini is the 4B member of the Vela 3.0 family, built on d3-mini. In one `system_one` call it answers routing and general decisions and finds the exact spans you name (entities, relations, personal data,

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89

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Zero-Shot Classification
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Model tree for vllm-sr/vela3-mini
Base model Qwen/Qwen3.5-4B-Base
Finetuned Qwen/Qwen3.5-4B
Finetuned vllm-sr/d3-mini
Adapter (1) this model

Collection including vllm-sr/vela...
Vela 3.0 Collection
Six open routin... • 7 items • U... • △ 7

unsupported claims, evidence), with probabilities and character offsets and no generated text.

Parameters	4.54B with vision encoder (base) + 973 MB of adapters and heads
Base	vllm-sr/d3-mini at revision `566da0ea`, unchanged
Inputs	Text or JSON, plus images
Answers	Choice · Yes / No · Score · Span
Licence	Apache-2.0

Summary

vela3-mini is the balance point of the Vela 3.0 family. d3-mini, frozen, answers every ordinary decision, bit for bit, and gated adapters add the spans you name, entities, relations and router signals. At 4B it is at or above every model the GLiFormer paper reports on classification and relations, and it writes its own definition for each label you name.

Highlights

- **Zero-shot classification:** 80.2 on the 13-set GLiFormer classification suite, 0.4 above GPT-5-mini: 79.8 (reported, GLiFormer paper Tab 5).
- **Relations:** 30.1 end to end (GLiFormer Table 4 protocol, avg of 4), above every model reported (best: Qwen3.5-9B, prompted, 25.7; reported, GLiFormer paper Tab 4).
- **Zero-shot NER:** 65.7 on OOD-7 with its own label definitions (label names in; written at

inference and cached per label set); 63.3 from label names alone.

- **Evidence:** 45.3 word-F1 on QASPER, zero-shot, against Vela 2.0 4B: 31.5 (same harness).
- **Router signals:** better than Vela 2.0 4B on 18 of the 24 test files of the public router signal suite, level on 5, worse on 1 (paired 95% CI).
- **Decisions, bit for bit:** every ordinary decision is answered by `d3-mini@566da0ea` itself, adapters off (561 of 561 identical in our check).



vela3-mini (4B) against reported numbers. Accent = vela3-mini; grey = other models (reported).

At a glance

What it is	A 4.5B decision-and-extraction model for LLM routing and guardrails. It runs vllm-sr/d3-mini, a mid-size d3 decision model, unchanged, and adds the exact spans you name: entities, relations, personal data, unsupported claims and evidence.
How it was trained	d3-mini stays frozen, so every ordinary decision is d3-mini's own answer, bit for bit. Vela adds small gated LoRA adapters on top, one per job: spans (unsupported claims,

	personal data, evidence), entities, relations, router signals and image hallucination spans. Each adapter switches on only for its own questions. Router signals are trained supervised, as in Vela 2.0, on the public router datasets Vela 2.0 used (for example AEGIS 2.0, PolyGuardMix and Nemotron-Safety-Guard v3 for safety, Global-MMLU for domain, WildFeedback for feedback, DiffusionDB for modality, MASSIVE for intents); domain and feedback also on the public router signal suite's train split. Other training data: public NER, relation, hallucination, evidence and PII datasets, plus web and document text labelled by a large teacher model. No test split and no Decision Index item is trained on; some router-signal-suite test rows also occur in public training sources, so the router table also reports the rows without that overlap.
What it is good at	Zero-shot classification; zero-shot relations; zero-shot NER; router signals above Vela 2.0 4B on 6 of 8 signals of the public router signal suite. Pick it for a balance of accuracy and speed.
How we measure it	Public test sets only, run through this package one request at a time with the shipped thresholds: CoNLL04, CoNLL04 · DocRED · FewRel · CrossRE (avg), OOD-7 (CrossNER + MIT), RAGTruth, Super-NI task 1568, the 13-set GLiFormer classification suite, the Decision Index 0.4.1, the LettuceDetect v2 test set, the public router signal suite. Other models' numbers are as their authors report them on the same test set and metric, or scored by us on the same rows where a table says so.

What it does

Capability	Ask (one question of a `system_one` call)	Get
Decisions	`{"type": "choice", "instructions": "Which topic?", "criteria": {"billing": None, "tech": None}}` (also `noul`, `score`)	a probability per option, from d3-mini itself
Router signals	a built-in router question id from `router_questions.json`	the signal's answer from the gated decision adapter
Named entities	`{"type": "span", "criteria": ["drug", "dosage"]}`	spans with label, offsets and score
Relations	`{"type": "span", "instructions": "Which spans are the founder of Apple?", "criteria": {"founder": None}}`	the related spans
Personal data	`{"type": "span", "criteria": ["PERSON", "EMAIL_ADDRESS"]}`	spans of the 17 trained PII types, any subset
Unsupported claims	state `{"context": ..., "answer": ...}` and `{"type": "span", "criteria": {"unsupported": None}}`	the words of the answer the context

Capability	Ask (one question of a system_one call)	Get
		does not support
Evidence	`{"type": "span", "criteria": ["evidence"]}`	the sentences that support the answer
Image-grounded claims	the unsupported-claims question with `images=["photo.jpg"]`	the words the image does not support

Quick start

```
# pip install "transformers==5.17.0" torch t
from transformers import AutoModel

m = AutoModel.from_pretrained("vllm-sr/vela3

# Name any labels; the spans come back with
r = m.system_one(
    state="Geoffrey Hinton presented capsule
    questions={"ents": {"type": "span", "cri
# researcher "Geoffrey Hinton", algorithm "c

# Unsupported claims in an answer, against a
r = m.system_one(
    state={"context": "The Eiffel Tower is 3
           "answer": "The Eiffel Tower is 33
    questions={"halu": {"type": "span", "cri
# flags "1925 by Gustave Eiffel's son"; add

# PII, relations and evidence use the same c
r = m.system_one(
    state="Steve Jobs founded Apple in Cuper
```

```
    questions={"pii": {"type": "span", "crit
               "rel": {"type": "span", "inst
)

# Any other choice / yes-no / score question
```

Evaluation

All numbers on public test sets, measured through this package with `system_one`, one request at a time and the shipped thresholds. Other models' numbers are as reported by their authors (or scored in the same harness where marked). Full per-set tables: EVALUATION.md.

▼ Named entities

Benchmark	vela3-mini	Best other model (reported)	Vela 2.0 4B	Size-class best (3-5B, reported)
OOD-7, label names only (F1)	63.3	B2NER-20B 72.1 (standardised label names; B2NER Tab 3)	40.7	JPT-4B, label names 58.3 (Just Pass Twice, Table 4)
OOD-7, label names in, own definitions (default)	65.7	same	same	same
NER, 26 datasets of the GLiFormer	65.0	–	–	–

Benchmark	vela3-mini	Best other model (reported)	Vela 2.0 4B	Size-class best (3-5B, reported)
paper (default mode; not all zero-shot)				

Label names are the only input; the model writes its own definitions at inference (d3-mini itself, adapters off) and caches them per label set. Training holds no CrossNER or MIT text, though many generic CrossNER type names occur as training labels.

▼ **Relations**

Benchmark	vela3-mini	Best other model (reported)	Vela 2.0 4B	Size-class best (3-5B, reported)
CoNLL04, head entity given, label names (F1)	**42.5**	GLiFormer-base 8.6 (same harness, Vela 2.0 card)	9.6	–
CoNLL04, head entity given, own definitions (default)	**47.6**	same	same	–

Benchmark	vela3-mini	Best other model (reported)	Vela 2.0 4B	Size-class best (3-5B, reported)
End to end, GLiFormer Table 4 protocol, avg of 4 (strict F1)	**30.1**	Qwen3.5-9B, prompted 25.7 (GLiFormer paper Tab 4)	–	no 3-5B entry; GLiNER-Relex, size not given 25.6
End to end: CoNLL04 · DocRED · FewRel · CrossRE (CoNLL04 zero-shot, others in-domain)	38.2 · 27.9 · 31.6 · 22.5	GPT-5-mini 42.4 · GLiNER-Relex 31.3 · Qwen3.5-9B 71.6 · GLiNER-Relex 18.1 (GLiFormer paper Tab 4)	–	–

CrossRE 22.5 (in-domain) is above every model reported in the GLiFormer paper (best: GLiNER-Relex 18.1).

▼ **Evidence**

Benchmark	vela3-mini	Best other model (reported)	Vela 2.0 4B	Size-class best (3 5B, reporte
QASPER, word-F1 (zero-shot)	**45.3**	–	31.5 (same harness)	–
ACL-Verbatim, word-F1 (in-domain)	41.5	ACL-Verbatim ModernBERT 53.4 (in-domain, evidence only; Vela 2.0 card)	24.4	–

ACL-Verbatim is in-domain: its silver training split is in the training data.

▼ **Unsupported claims (hallucination spans)**

Benchmark	vela3-mini	Best other model (reported)	Vela 2.0 4B	Siz cla best 5E repor
LettuceDetect v2 test set, 10,698 answers (example-F1)	0.874	LettuceDetect v2 qwen-2b 0.921 (its card)	0.874	–
RAGTruth (example-F1)	0.775	HDM-2 (3B) 0.850 (HalluciNot Tab 2)	0.770	–

Benchmark	vela3-mini	Best other model (reported)	Vela 2.0 4B	Siz cla best 5E repor
M-HalDetect validation, image-grounded (char-F1)	0.575	–	no image path	–

▼ Zero-shot classification

Benchmark	vela3-mini	Best other model (reported)	Vela 2.0 4B	Size-class best (3-5B, reported)
GLiFormer suite, 13 sets, label names only (mean macro-F1)	80.2	GPT-5-mini 79.8 (GLiFormer paper Tab 5)	–	no 3-5B entry; GPT-5-mini, size not given 79.8

These questions run on d3-mini with every adapter off.

▼ **Router signals** (public router signal suite; above Vela 2.0 4B on 6 of 8)

Signal (metric, test rows)	vela3-mini	Vela 2.0 4B, same rows	Decision-1.0-Route-0.6B, same rows	Rows sharing no text with training: vela3-mini / Vela 2.0 4B
Harmful request (AUC, 2,230)	0.952	0.956	0.963	0.942 / 0.945 (1,254)
Jailbreak or prompt injection (AUC, 1,878)	**0.967**	0.913	0.991	**0.966** / 0.862 (944)
Personal data in the request (AUC, 3,000)	**0.956**	0.943	0.994	**0.958** / 0.941 (2,802)
Answer not supported by its context (AUC, 3,000)	0.763	0.757	0.791	**0.762** / 0.733 (1,743)
Fact-check needed (AUC, 800)	**0.831**	0.799	0.942	**0.820** / 0.800 (690)
Output modality (text or image) (AUC, 3,000)	**0.957**	0.900	0.999	**0.955** / 0.895 (2,916)
Domain (subject area)	**0.872**	0.757	0.863	**0.873** / 0.758 (2,916)

Signal (metric, test rows)	vela3-mini	Vela 2.0 4B, same rows	Decision-1.0-Route-0.6B, same rows	Rows sharing no text with training: vela3-mini / Vela 2.0 4B
(accuracy, 2,996)				
User feedback on the previous answer (accuracy, 2,842)	**0.835**	0.613	0.848	**0.819** / 0.573 (2,295)

vllm-sr/router-signal-suite (public), the `test` file of each signal, scored with the dataset's own scoring code. Each signal is a router question this package was trained on (supervised, as in Vela 2.0), asked one request at a time with the shipped thresholds.

Vela 2.0 4B was run by us through its public release on the same rows. Decision-1.0-Route-0.6B is scored from the suite's published predictions; it was trained on the suite's own train split, so these test files are in-distribution for it.

Bold: the paired 95% bootstrap CI against Vela 2.0 excludes a tie. The last column drops the test rows that share text (exact or a 13-gram) with this model's training data. Safety and jailbreak are scored on the rows whose text the suite publishes (2,230 of 3,003 and 1,878 of 3,000).

Domain and feedback are also trained on the suite's own train split (with the public router data Vela 2.0 used; the suite's test and hold rows, and anything sharing an exact text or a 13-gram with them, are kept out), as Decision-1.0-Route is, so their test files are in-distribution for both. The domain question has a 15th option, religion, which the suite maps to philosophy. Feedback has NO_FEEDBACK as a trained option.

▼ **Decisions** (general decisions are the base model's, bit for bit)

Decision Index 0.4.1 (2026-10-28)	Public index	Full score (board rank)
Fastino GLiDE 1.2 (tied rank)	71.99	71.52 (1)
StartLux-Decision v1.1 4B (tied rank)	67.81	54.80 (73)
Perplexity Decider Flash v1.2 (4B) (tied rank)	60.06	54.29 (75)
d3-mini, the base's board entry (this package pins revision 566da0ea)	54.90	49.97 (87)
OneJev1.6-4B (tied rank)	61.95	48.88 (90)
GLiClass v4 Edge (0.4B) (tied rank)	47.49	31.21 (156)
OneJev1.6-0.8B (tied rank)	48.76	29.58 (162)
StartLux-Decision v1.1 0.8B (tied rank)	53.79	28.88 (167)

General decisions are the base model's own answers, bit for bit, so the base's board entry is this

package's decision score. Full score (the board's ranking): 60% private same-skill tests and 40% private new-domain tasks; public index: the 37 public benchmarks, chance-corrected. All values are board entries on the Decision Index 0.4.1 board (multimodalart/jev-decision-index, data generated 2026-10-28T09:02Z) unless noted; the rank is the position in the board's Full-score order, and tied rank marks entries the board shows as tied. On the board's Full score, d3-mini scores 49.97 (rank 87, tied). This package pins d3-mini revision `566da0ea`; the board does not state which d3-mini revision it ran.

Benchmark	vela3-mini	Best other model (reported)	Vela 2.0 4B	Size-class best (3-5B, reported
Super-NI task 1568, held out (accuracy)	0.611	–	0.494	–
MASSIVE, three unseen scenarios: intent / scenario (accuracy; 14,911 decontaminated rows)	0.778 / 0.890	–	0.727 / 0.814 (full set)	–

MASSIVE: three scenarios never used in our router training (iot, transport, cooking), 52 locales; accuracy on the 14,911 test utterances that match no training text of the model's lineage. The 342 excluded utterances occur verbatim in upstream

training mixtures (tasksource MASSIVE / HWU64 rows). Vela 2.0 values are on the full 15,253, so these rows are not bolded. Super-NI 1568: 79 to 90 of its 162 ProPara passages share a 13-gram with other tasks' training rows.

▶ **Latency**

Benchmarks

- **OOD-7** (CrossNER AI, literature, music, politics, science; MIT movie, restaurant): find the entities of the named types in text from domains the model was not trained on (strict entity F1).
- **NER, 26 datasets of the GLiFormer paper** (IE_INSTRUCTIONS): entities of named types across 26 public NER sets, not all zero-shot (strict entity F1).
- **CoNLL04, head entity given:** given one entity of a sentence, find the entities it is related to by the named relation (F1 over those tail entities).
- **Relations end to end, GLiFormer Table 4 protocol** (CoNLL04, DocRED, FewRel, CrossRE): find the entity pairs and their relation from raw text (strict triplet F1).
- **QASPER:** find the sentences of a research paper that answer a question about it (word-F1).
- **ACL-Verbatim** (KRLabsOrg/acl-verbatim-spans): find the passage of an ACL paper that supports an answer (word-F1).

- **M-HalDetect** (validation): which words of an image description the image does not support (char-F1).
- **LettuceDetect v2 test set** (10,698 answers, as in the LettuceDetect v2 paper (GroundLM 2026 workshop at EMNLP 2026) and card): does an answer contain claims its context does not support (example-F1: the answer is flagged or not).
- **RAGTruth:** does a retrieval-augmented answer contain claims its retrieved context does not support (example-F1).
- **MASSIVE, three unseen scenarios** (iot, transport, cooking; 52 languages): pick the intent and the scenario of a user command the model never saw in training (accuracy).
- **Super-NI task 1568** (ProPara): answer a multiple-choice question about how a process changes an entity (accuracy).
- **Zero-shot classification, 13 sets** (the GLiFormer paper's suite: SST-2, IMDB, AG News, Banking77, and more): pick the label of a text from label names only (mean macro-F1).
- **Decision Index 0.4.1** (board, 2026-10-28): 37 public decision benchmarks plus private tests; the Full score ranks on the private same-skill and new-domain tests (chance-corrected skill, 0 = guessing).
- **Router signal suite** (vllm-sr/router-signal-suite): the vLLM Semantic Router's public test of its request signals (harmful requests, jailbreaks, personal data, unsupported answers, fact-check need, modality, domain,

feedback); each label comes from public corpora (AUC or accuracy).

The family



Model	Size	Base	Best for
vela3-fast	0.4B	UltraBERT-380M encoder	the default router; fastest: decisions + spans in one ~14 ms encoder pass, English, 32K
vela3-fast-vision	0.3B	UltraVBERT encoder	image decisions, image-grounded hallucination spans, region pointing (preview)
vela3-lite	0.8B	d3-lite	small GPUs, low latency
vela3-mini	4B	d3-mini	balanced accuracy and speed
vela3-flash	9B	d3-flash	high-accuracy routing
vela3-turbo	27B	d3-turbo	the reference model

How it works

- **Frozen base.** `d3-mini@566da0ea` runs unchanged; ordinary questions are its own answers.

System theme TOS Privacy About Careers Models Datasets Spaces Pricing Docs

- **Gated LoRA adapters.** Each switches on only for its own questions, and every answer names the one it used (`head_reason`): decision (router (entity labels with definitions), relation, image (hallucination spans with images), span (evidence, PII, unsupported claims, toxic spans, names-only labels), and an optional decision adapter for any other question (`"decision_adapter": true`).
- **Spans as decisions.** The labels are presented as options; a span head scores every word against every label, and agreeing neighbours form a span with character offsets.
- **Self-written definitions.** Before reading, the model (d3-mini itself, adapters off) writes a one-line definition for each undescribed label, cached per label set: OOD-7 goes from 63.3 (names) to 65.7.
- **Specification.** Adds 973 MB of adapters and heads; image threshold 0.55; text or JSON state, images and video for decisions, images for hallucination spans. Full evaluation: EVALUATION.md.

Trade-offs

- **Unsupported claims:** hallucination spans 0.874 against 0.921 for LettuceDetect v2 qwen-2b, RAGTruth 0.775 against 0.850 for HDM-2 (3B) (reported).
- **General decisions are d3-mini's:** its board entry is below StartLux-Decision v1.1 4B (full score 54.80 vs 49.97) and Perplexity Decider Flash v1.2

(4B) (full score 54.29 vs 49.97) on the Decision Index 0.4.1.

- **NER from label names:** 63.3 on OOD-7, level with the best reported encoder (gliner-multitask-v1.0 63.25) and below GNER-T5-xl (3B) 66.1 (reported).

Citation

```
@misc{vela3_mini,
  title        = {vela3-mini (Vela 3.0, 4B):
  author       = {{KR Labs} and {vLLM Semant
  year         = {2026},
  howpublished = {\url{https://huggingface.c
}
```